

Mock EdTech Dataset

SCENARIO

You are a Data Analyst at LearnSphere, a South African online learning platform offering courses in tech, business, and design skills. Management wants a clearer picture of student growth, course performance, revenue, and engagement ahead of the next quarterly planning meeting. The data was exported from three different internal systems (CRM, LMS, and Payments) and has NOT been cleaned, duplicates, inconsistent formats, missing values and a few errors are present, just like a real export would be.

YOUR TASK

1. Clean the data using SQL or Excel.
2. Build a data model linking the 5 tables (see schema below).
3. Build a Power BI dashboard answering the business questions below.

TABLES PROVIDED

Students - one row per student, signup details (PK: student_id)
Courses - course catalog, pricing, category, instructor (PK: course_id)
Instructors - instructor bios and ratings (PK: instructor_id)
Enrollments - one row per student-course enrollment, status, payment (PK: enrollment_id)
Reviews - student reviews tied to an enrollment (PK: review_id)

RELATIONSHIPS

Enrollments.student_id -> Students.student_id
Enrollments.course_id -> Courses.course_id
Courses.instructor_id -> Instructors.instructor_id
Reviews.enrollment_id -> Enrollments.enrollment_id

KNOWN DATA QUALITY ISSUES TO CLEAN

- Duplicate student and enrollment records
- Inconsistent date formats (YYYY-MM-DD, DD/MM/YYYY, DD-Mon-YYYY)
- Inconsistent text casing (e.g. 'completed' vs 'COMPLETED', city names, course titles)
- Inconsistent ID casing (e.g. 'STU0012' vs 'stu0012')

- Missing values in email, phone, age, price, completion_pct, ratings, etc.
- Invalid values (negative age, negative amount_paid_zar)
- Inconsistent categorical values (Yes/No/Y/N/TRUE/FALSE for is_active)
- Inconsistent phone number formats
- Trailing/leading whitespace in some text fields
- Boolean-like and free-text fields mixed (status, device_type, signup_source)

BUSINESS QUESTIONS TO ANSWER

1. How has student signups and enrollment volume trended over the last 3 years?
2. Which course categories generate the most revenue, and which have the highest dropout rate?
3. What is the course completion rate overall, and which courses/instructors underperform?
4. Which signup source and device type bring in the most paying, completing students?
5. Which cities/provinces should GoLearn target for marketing spend, based on enrollment and revenue?
6. What is the average student rating per course/instructor, and does it correlate with completion rate?
7. What % of enrollments used a discount (amount paid below course price), and how does that affect completion?
8. Who are the top 10 instructors by revenue generated and by average rating?

SUGGESTED DELIVERABLE

A Power BI dashboard (2-3 pages) covering: Overview KPIs, Course/Category performance, Student & Marketing insights, Instructor performance, plus a short written summary of the data cleaning steps you took and why.